

Evaluating Adaptive Timestep Sampling for the Efficient Training of Flow Matching Models

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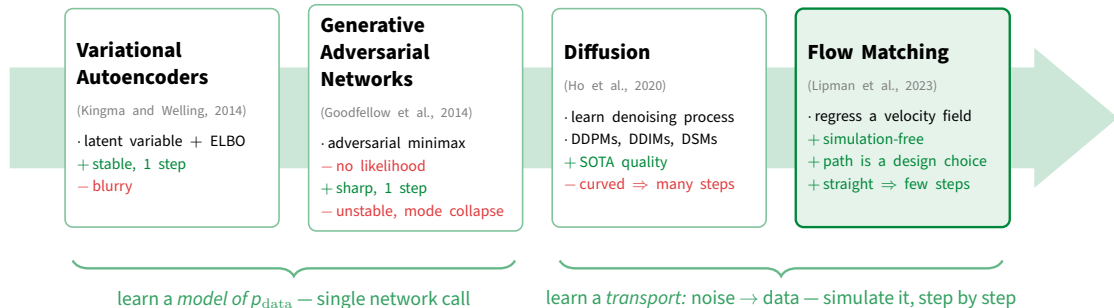
noise $t = 0$

$x_t = (1 - t)x_0 + tx_1$

$t = 1$ generated samples

- 1 Introduction to Flow Matching
- 2 Related Work on Training Efficiency
- 3 Adaptive Timestep Sampling for Flow Matching
- 4 Experiments
- 5 Conclusion

“Creating noise from data is easy; creating data from noise is generative modeling”
— Song et al., 2021



A CNF is an ODE [1]

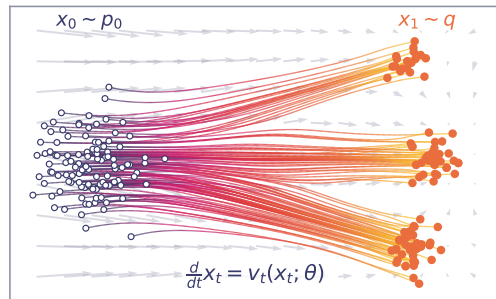
- **Flow** ϕ_t transports the prior to data

$$\phi_t(x_0) = x_0 + \int_0^t v_s(\phi_s(x_0), s) ds, \quad p_t = [\phi_t]_* p_0$$

- **ODE** describes the flow...

$$\frac{d}{dt}x_t = v_t(x_t; \theta), \quad x_0 \sim p_0 = \mathcal{N}(0, I)$$

- ...and defines a **velocity field** $v_t(x_t; \theta)$ parametrized by a neural network with parameters θ
- Arbitrary probability paths possible
⇒ Continuity equation: Conservation of probability mass at every t



The learned field (grey, at $t=0.5$) carries the prior on the left onto the data modes on the right.

1. Regress the field, not the likelihood [17]

$$\mathcal{L}_{\text{FM}}(\theta) = \mathbb{E}_{t, x_t \sim p_t} \|v_t(x_t; \theta) - u_t(x_t)\|^2$$

- **But:** the marginal target is intractable

$$u_t(x_t) = \int u_t(x_t|x_1) \frac{p_t(x_t|x_1)q(x_1)}{p_t(x_t)} dx_1$$

2. Marginalization trick $\Rightarrow$ Conditional FM

$$\mathcal{L}_{\text{CFM}}(\theta) = \mathbb{E}_{t, x_1 \sim q, x_t \sim p_t(\cdot|x_1)} \|v_t(x_t; \theta) - u_t(x_t|x_1)\|^2$$

- Pick a tractable *per-sample* path $p_t(x_t|x_1)$
- Averaging the conditional fields provably generates the marginal path
- Identical gradients $\Rightarrow \nabla_{\theta} \mathcal{L}_{\text{FM}} = \nabla_{\theta} \mathcal{L}_{\text{CFM}}$

details

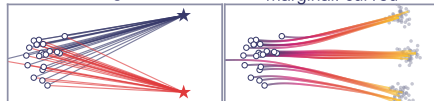
3. Choice of probability path

details

Linear interpolation (CondOT):

$$x_t = (1-t)x_0 + tx_1, \quad \Rightarrow u_t = x_1 - x_0$$

conditional: straight, $u_t = x_1 - x_0$ marginal: curved



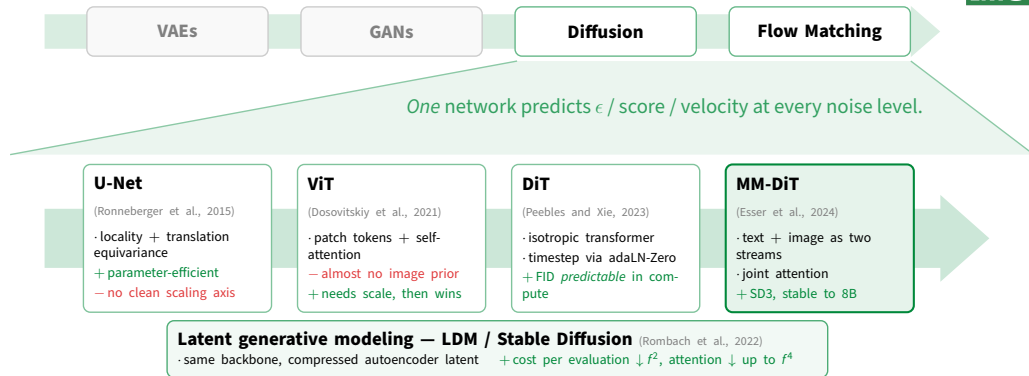
4. Inference: Integrating the ODE

Euler: $x_{t+\Delta t} = x_t + \Delta t v_t$, error $\mathcal{O}(\Delta t^2)$



straighter $\Rightarrow$ fewer NFE [11]

What Parameterizes v_t : The Backbone Architectures



Training efficiency has become central concern

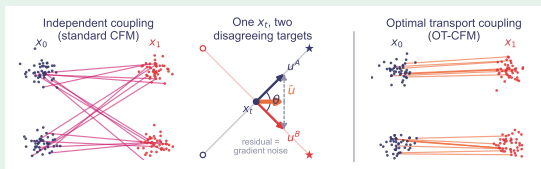
- **Inference is largely solved:** reflow + distillation $\rightarrow$ a handful of passes, ultimately one [18, 26]
- **Model size keeps growing:** SD3 (8B) up to 5×10^{22} FLOPs [4]; Movie Gen 30B [22]

Outline

- 1 Introduction to Flow Matching
- 2 Related Work on Training Efficiency**
- 3 Adaptive Timestep Sampling for Flow Matching
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I. Optimize Boundary Couplings

details



Mini-batch OT [28], multisample couplings [23], immiscible diffusion [16], augmented FM [33], stable target field [32]

→ which x_0 is paired with which x_1

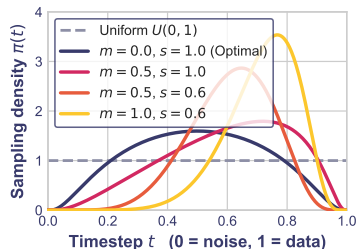
II. Shift the Optimization Focus Across Timesteps

details

$$\mathcal{L}_{\text{CFM}}(\theta) = \mathbb{E}_{t \sim \pi(t), x_1, x_0} \left[w(t) \left\| v_t(\alpha_t x_1 + \beta_t x_0; \theta) - u_t(\cdot | x_1) \right\|^2 \right]$$

- ✗ **Noise schedule** (α_t, β_t) — locked by the linear interpolant; changing it re-curves the paths
 - ✗ **Loss weighting** $w(t)$ — only rescales gradients, without shifting where compute is spent
 - ✓ **Timestep sampling** $\pi(t)$ — changes the *batch composition*, so it can actually reallocate compute
- where along the path compute is spent

Logit-normal (Esser et al., 2024)

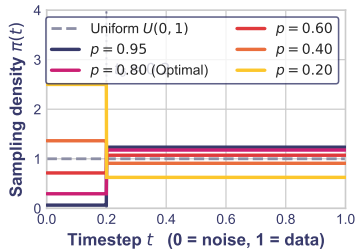


$$t = \sigma(z), z \sim \mathcal{N}(m, s^2)$$

Rationale: the boundaries are trivial (prior mean / data mean); the interesting dynamics sit at $t \approx 0.5$.

Best: $m=0, s=1$ — the SD3 default, beats uniform at every step count.

Bernoulli (Zheng et al., 2024)

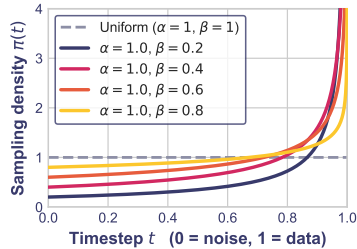


Piecewise-uniform: mass p on $(t_\delta, 1]$, $1 - p$ on $[0, t_\delta]$.

Rationale: the W_2 distance between marginals decays exponentially $\Rightarrow$ the largest change sits near the data.

Best: $t_\delta=0.2, p=0.8$ (DDPM-BS) — faster convergence, better FID.

Beta (Zheng et al., 2024)



$$f(t; \alpha, \beta) \propto t^{\alpha-1}(1-t)^{\beta-1}.$$

Rationale: $\text{KL}(p_t \| \mathcal{N}(0, I))$ decays exponentially away from the data $\Rightarrow$ J-shaped density.

Best: $\alpha=1, \beta=0.8$ (B-TTDM) — up to $4.71\times$ faster on ImageNet.

Two observations on DDPM training

- 1 Per-timestep **difficulty** (gradient variance) spans orders of magnitude
- 2 $\mathcal{L}_t$ are **interdependent**: oversampling one t -regime degrades the fit elsewhere

⇒ Learn $\pi(t)$ online

Central Research Question

- For DDPMs: ATS leads to faster convergence and improved FID
- ⇒ Do these benefits transfer to Flow Matching with CondOT paths?

- 1 Introduction to Flow Matching
- 2 Related Work on Training Efficiency
- 3 Adaptive Timestep Sampling for Flow Matching
 - Adaptive Timestep Sampling (ATS)
 - Advantage Actor-Critic ATS (A2C-ATS)
 - Curvature-Aware ATS (LTE-ATS)
- 4 Experiments
- 5 Conclusion

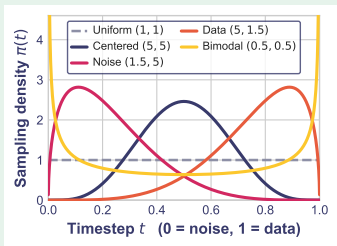
Idea

Sample t that lead to *highest expected loss reduction* $\Rightarrow$ re-evaluate continuously

Sample t from adaptive Beta

Policy network $\pi_\phi(\cdot|x_1)$ parametrizes cond. Beta:

$$\alpha_i, \beta_i = \pi_\phi(\cdot|x_1^{(i)}), \quad t_i \sim \text{Beta}(\alpha_i, \beta_i)$$



Train $\pi_\phi(\cdot|x_1)$ to maximize the expected loss reduction

$$\phi^* = \arg \max_{\phi} \mathbb{E}_{x_1, x_0, t \sim \pi_\phi(\cdot|x_1)} [\Delta_k^t]$$

- Exp. loss reduction when updating at t :

$$\theta_{k+1} = \theta_k - \gamma \nabla_{\theta_k} \mathcal{L}(x_1, t)$$

$$\Delta_k^t = \int_0^1 \underbrace{(\mathcal{L}_{\theta_k}(x_1, \tau) - \mathcal{L}_{\theta_{k+1}}(x_1, \tau))}_{\delta_{k,\tau}^t} d\tau$$

- $\Rightarrow$ Computationally infeasible
- $\Rightarrow$ Approximate loss reduction Δ_k^t with best explaining proxy timesteps $\tau' \in S$

The Policy Objective: Global Loss Reduction Δ_k^t

Approximation of global loss reduction Δ_k^t by best explaining proxy timesteps τ'

$$\Delta_k^t = \int_0^1 \underbrace{(\mathcal{L}_{\theta_k}(x_1, \tau) - \mathcal{L}_{\theta_{k+1}}(x_1, \tau))}_{\delta_{k,\tau}^t} d\tau, \quad \theta_{k+1} = \theta_k - \gamma \nabla_{\theta_k} \mathcal{L}(x_1, t)$$

- 1 Select $|S| := 3 \ll T$ proxy timesteps τ' by online F-regression, $F_\tau = \frac{r_\tau^2}{1-r_\tau^2} (|Q| - 2)$
- 2 Evaluate δ only at those $\tau' \in S$, then average:

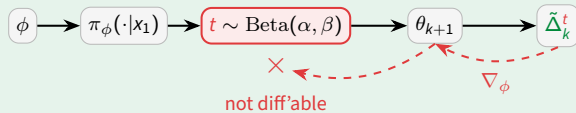
$$\tilde{\Delta}_k^t = \frac{1}{|S|} \sum_{\tau' \in S} \delta_{k,\tau'}^t \approx \Delta_k^t$$

Rolling queue Q picks the τ that best explain the global reduction; policy updated every f_S backbone steps.

Plug the proxy into the policy objective $\Rightarrow$ no gradient

$$\phi^* = \arg \max_{\phi} \mathbb{E}_{x_1, x_0, t \sim \pi_{\phi}(\cdot | x_1)} [\tilde{\Delta}_k^t]$$

t is drawn from $\pi_{\phi} \Rightarrow t$ depends on ϕ



REINFORCE (Williams, 1992)

- Policy $\pi_\phi(a|s)$ samples an action a in state s ; reward R follows

→ Objective: $J(\phi) = \mathbb{E}_{a \sim \pi_\phi}[R]$

- Log-derivative trick moves ∇_ϕ inside the expectation:

$$\nabla_\phi J = \mathbb{E}_{a \sim \pi_\phi} \left[R \underbrace{\nabla_\phi \log \pi_\phi(a|s)}_{\text{eligibility } \mathbf{e}} \right]$$

⇒ Estimate gradient via Monte Carlo

- Update $\Delta\phi = \alpha (R - b) \mathbf{e}$ with optional baseline b

Theorem (Williams, 1992)

Proof

$$\mathbb{E}[\Delta\phi | \phi] = \alpha \nabla_\phi \mathbb{E}[R | \phi]$$

- In expectation: REINFORCE update = true gradient

⇒ unbiased estimator

The REINFORCE components in ATS

policy $\pi_\phi(a|s) \mapsto \pi_\phi(t|x_1) = \text{Beta}(\alpha_i, \beta_i)$

state $s \mapsto$ clean image x_1

action $a \mapsto$ timestep t

reward $R \mapsto \tilde{\Delta}_k^t$ (proxy loss reduction)

eligibility $\mathbf{e} \mapsto \nabla_{\phi_k} \log \pi_{\phi_k}(t|x_1)$

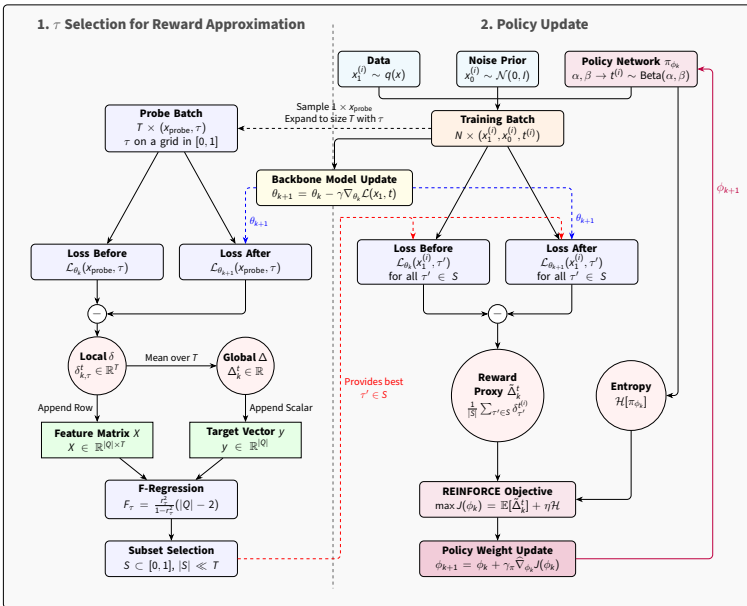
baseline $b \mapsto 0$

ATS policy update equation

$$\phi_{k+1} = \phi_k + \gamma_\pi \left(\underbrace{\tilde{\Delta}_k^t \nabla_{\phi_k} \log \pi_{\phi_k}(t|x_1)}_{\text{REINFORCE gradient } R \mathbf{e}} + \underbrace{\eta \nabla_{\phi_k} \mathcal{H}[\pi_{\phi_k}]}_{\text{entropy bonus}} \right)$$

- Entropy $\eta \mathcal{H}$ prevents premature Beta collapse

The Full ATS Process



Problem

REINFORCE gradients are inherently noisy
⇒ Slows down policy adaptation [5]

Proposed Solution

Center the reward by subtracting a baseline
⇒ Optimize **advantage**: $A = R - b$

Value network (critic) as baseline

- $V_\psi(x_1)$ predicts the expected reward of an image, marginalizing t and x_0

$$\mathcal{L}_{\text{value}}(\psi) = \frac{1}{B} \sum_i (V_\psi(x_1^{(i)}) - \tilde{\Delta}_k^{t(i)})^2$$

if successful $\rightarrow V_\psi(x_1) \approx \mathbb{E}_{t, x_0} [\tilde{\Delta}_k^t | x_1]$

- Advantage $A = R - V = \tilde{\Delta}_k^t - V_\psi(x_1)$
- $A > 0$: better than this image's average ⇒ push up t -density
- $A < 0$ ⇒ push down t -density

A2C-ATS update

$$\phi_{k+1} = \phi_k + \gamma_\pi \left(\underbrace{(\tilde{\Delta}_k^t - V_\psi(x_1))}_A \nabla_{\phi_k} \log \pi_{\phi_k}(t|x_1) + \eta \nabla_{\phi_k} \mathcal{H}[\pi_{\phi_k}] \right)$$

Policy gradient comparison

$$\hat{g}_{\text{pg}}^{\text{ATS}} = R \nabla_{\phi_k} \log \pi_{\phi_k}(t|x_1) \quad \text{vs.} \quad \hat{g}_{\text{pg}}^{\text{A2C}} = (R - V) \nabla_{\phi_k} \log \pi_{\phi_k}(t|x_1)$$

reward $R = \tilde{\Delta}_k^t$, baseline $V = V_\psi(x_1)$

1. A2C introduces zero bias

$$\begin{aligned} & \mathbb{E}_{t \sim \pi_{\phi_k}} \left[V_\psi(x_1) \nabla_{\phi_k} \log \pi_{\phi_k}(t|x_1) \right] \\ &= V_\psi(x_1) \int \pi_{\phi_k}(t|x_1) \nabla_{\phi_k} \log \pi_{\phi_k}(t|x_1) dt \\ &= V_\psi(x_1) \int \nabla_{\phi_k} \pi_{\phi_k}(t|x_1) dt \\ &= V_\psi(x_1) \nabla_{\phi_k} \underbrace{\left(\int_0^1 \pi_{\phi_k}(t|x_1) dt \right)}_{=1} = 0 \end{aligned}$$

$$\Rightarrow \mathbb{E}[\hat{g}_{\text{pg}}^{\text{A2C}}] = \mathbb{E}[\hat{g}_{\text{pg}}^{\text{ATS}}] - \text{unbiased}$$

2. A2C reduces gradient variance

details

Difference in policy-gradient variance:

$$\begin{aligned} & \text{Tr Cov}(\hat{g}_{\text{pg}}^{\text{ATS}}) - \text{Tr Cov}(\hat{g}_{\text{pg}}^{\text{A2C}}) \\ &= \mathbb{E}[R^2 \|\nabla_{\phi_k} \log \pi_{\phi_k}\|^2] - \underbrace{\mathbb{E}[(R - V)^2 \|\nabla_{\phi_k} \log \pi_{\phi_k}\|^2]}_{\rightarrow 0 \text{ if } V \approx \mathbb{E}[R|x_1]} \end{aligned}$$

$$\Rightarrow \text{Tr Cov}(\hat{g}_{\text{pg}}^{\text{A2C}}) \ll \text{Tr Cov}(\hat{g}_{\text{pg}}^{\text{ATS}})$$

$\Rightarrow$ Policy adapts more precisely

Extension 2: Curvature-Aware ATS (LTE-ATS)

Idea: Focus on geometric bottlenecks

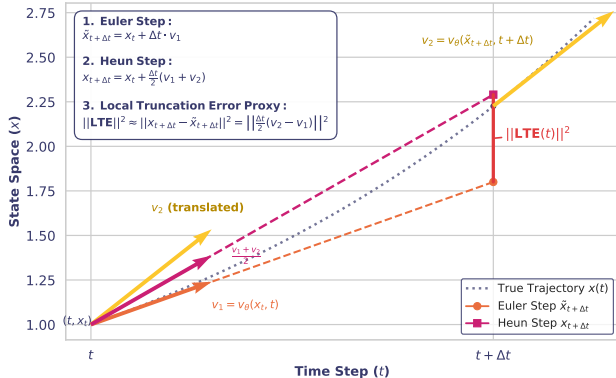
- Quality is limited by *integration error*
- ⇒ Oversample the timesteps hardest to integrate
- ⇒ $\pi(t) \propto \text{curvature}$

Curvature via Local Truncation Error

- Exact curvature intractable
- Approximate with 1st-order Euler vs. 2nd-order Heun step[10]

$$\widehat{\text{LTE}}_{\theta}(x_t, t) = \left\| \frac{\Delta t}{2} (v_{\theta}(\tilde{x}_{t+\Delta t}, t+\Delta t) - v_{\theta}(x_t, t)) \right\|_2^2$$

- Evaluated at each policy update on the current **training batch**
- Cost: **2 backbone forward passes**
⇒ cheaper than the ATS reward estimation



LTE-ATS update

$$\phi_{k+1} = \phi_k + \gamma \pi \left(\underbrace{\widehat{\text{LTE}}_{\theta}(z, t)}_{R = \text{curvature}} - V_{\psi}(x_1) \right) \nabla_{\phi_k} \log \pi_{\phi_k}(t|x_1) + \eta \nabla_{\phi_k} \mathcal{H}[\pi_{\phi_k}]$$

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- 4 **Experiments**
 - Setup
 - E1 – Generation Quality and Training Efficiency
 - E2 – Insights Into the Flow Matching Learning Process
 - E3 – Impact on Inference Trajectories
- 5 Conclusion

Experimental Setup – Does ATS Transfer to Flow Matching?

→ Fixed backbone, optimizer, and hyperparameters — only t -sampler varies.

Dataset and preprocessing



CIFAR-10 [15], 32×32 , unconditional — 50k train / 10k held-out, per-channel mean/std 0.5 $\Rightarrow x_1 \in [-1, 1]$.

U-Net backbone $v_\theta(x_t, t)$

architecture

- base width 128, multipliers (1, 2, 2, 2), 2 ResBlocks/stage
- self-attention at 16^2 and the 4^2 bottleneck, dropout 0.1
- sinusoidal t -embedding (t rescaled by 1000)
- $\approx 35.7\text{M}$ parameters — shared by *every* strategy

Policy network π_ϕ (critic V_ψ alike)

- conv encoder $3 \rightarrow 32 \rightarrow 64 \rightarrow 128$, pooled to 2048-d
- MLP 256, depth 2, outputs (α, β) of the Beta policy
- initialized at $\text{Beta}(1, 1) = \mathcal{U}[0, 1]$
- $\approx 0.68\text{M}$ parameters each (2% of the backbone)

Training configuration and hyperparameters

- Adam, lr 2×10^{-4} backbone / 10^{-2} actor / 10^{-3} critic
- 5k warmup, grad clip 1.0, EMA 0.9999, fp32
- 300 epochs, batch 256, $4 \times$ Titan X (DDP) = 58.5k steps
- CondOT $x_t = (1-t)x_0 + tx_1$, $u = x_1 - x_0$; Euler $N=50$

Experimental Setup - Backbone Architecture

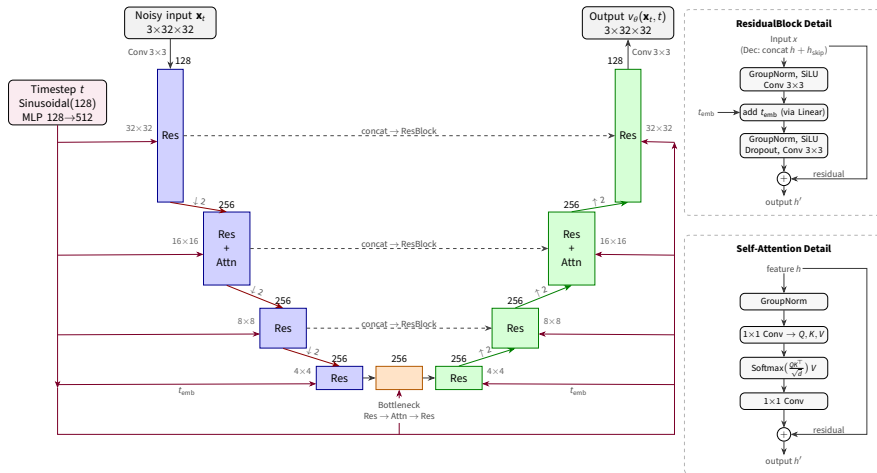


Figure: **Shared U-Net backbone**, $\approx 35.7\text{M}$ parameters, after Kim et al. [13]. Block *height* = spatial resolution ($32^2 \rightarrow 4^2$), *width* = channels ($128 \rightarrow 256$); dashed arrows are concatenating skips, the purple rail broadcasts t_{emb} into every residual block.

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Table: Timestep-sampling strategies evaluated in the comparison suite, together with the exact hyperparameter configurations tested across our runs.

Strategy	Sampling density	Tested configuration(s)	Reference
<i>Static samplers (fixed density)</i>			
Uniform	$\mathcal{U}(0, 1)$	baseline (no parameters)	Lipman et al. [17]
Logit-normal	$t = \sigma(z), z \sim \mathcal{N}(m, s^2)$	$m = 0, s = 1$	Esser et al. [4]
Bernoulli (data)	Piecewise: $f(t) \propto p$ on $(t_\delta, 1]$, else $1 - p$	$t_\delta=0.2, p \in \{0.8, 0.9\}$	Zheng et al. [34]
Bernoulli (noise)	Piecewise: $f(t) \propto p$ on $(t_\delta, 1]$, else $1 - p$	$t_\delta=0.2, p=0.2$	Zheng et al. [34]
Beta (data)	$\text{Beta}(\alpha, \beta)$	$(\alpha, \beta)=(1, 0.8)$	Zheng et al. [35]
Beta (noise)	$\text{Beta}(\alpha, \beta)$	$(\alpha, \beta)=(0.8, 1)$	Zheng et al. [35]
<i>Adaptive samplers (learned Beta policy, REINFORCE)</i>			
ATS	Learned Beta	$R = \Delta_k^t$	Kim et al. [12]
A2C-ATS	Learned Beta + critic	$R = \Delta_k^t - V_\psi(x_1)$	This work
LTE-ATS	Learned Beta + critic, LTE reward	$R = \widehat{\text{LTE}}_\theta(x_t, t) - V_\psi(x_1)$	This work

Experiment 1 – Generation Quality and Training Efficiency

Four hypotheses on one controlled CIFAR-10 suite

- H1.1** ATS leads to faster convergence than uniform. ▶ now
- H1.2 The A2C value baseline lowers the policy-gradient variance.
- H1.3 The curvature (LTE proxy) is a valid reward signal.
- H1.4 Adaptive timestep sampling outperforms static non-uniform sampling.

H1.1 – Charging the Cost of the Policy

H1.1: ATS leads to faster convergence than uniform.

Theoretical: $NFE_{\text{fwd-eq}}$

Backbone evaluations only, hardware-independent. A backward pass costs $\approx 3 \times$ a forward one:

$$NFE_{\text{fwd-eq}} = NFE_{\text{fwd}} + 3 NFE_{\text{bwd}}$$

- policy & critic excluded: 0.68M vs. 35.7M parameters

- Uniform sampling is effectively free
- ATS is more compute intense due to *reward estimation*.

Practical: Wall-clock time

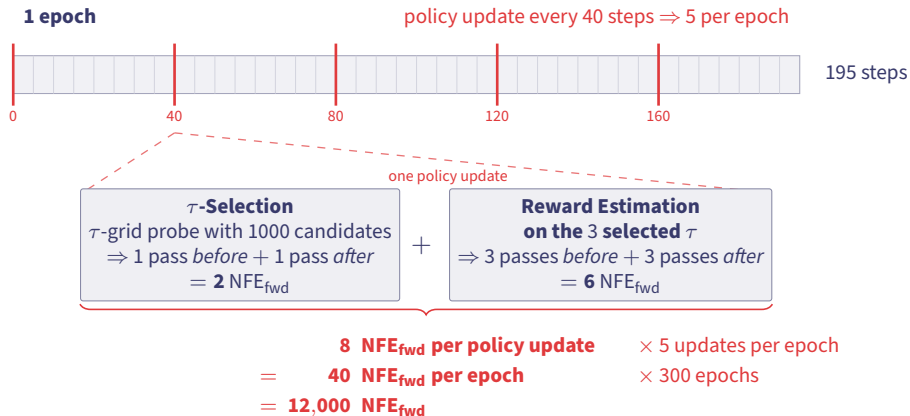
Measured on the fixed $4 \times$ Titan X setup. Captures what NFE cannot:

- actor and critic updates
- probe bookkeeping, feature selection

Hardware- and load-dependent $\Rightarrow$ a complement to NFE, not a replacement.

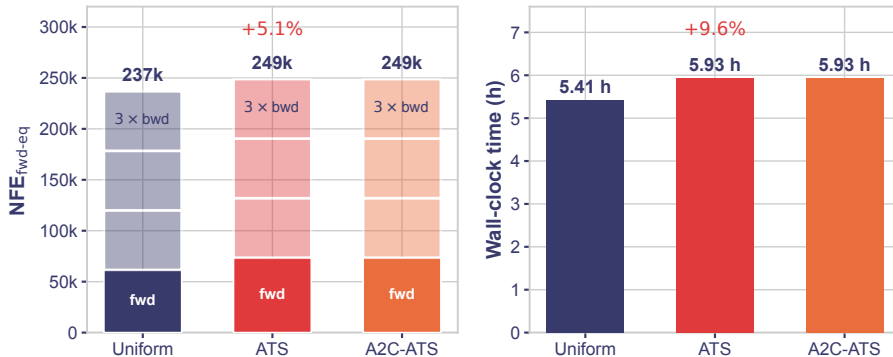
H1.1 – Where the ATS Overhead Comes From

H1.1: ATS leads to faster convergence than uniform.



H1.1 – The Overhead Is Real

H1.1: ATS leads to faster convergence than uniform.



Accounting for the computational overhead

- NFE_{fwd-eq} : +12k forward passes, 236,910 $\rightarrow$ 248,910 (+5.1%) $\Rightarrow$ independent of backbone size
- Wall-clock: 5.41 $\rightarrow$ 5.93 h (+9.6%) $\Rightarrow$ the untracked RL bookkeeping widens the gap

H1.1 – How We Measure Quality and Convergence

H1.1: ATS leads to faster convergence than uniform.

Sample quality: FID ↓

Fréchet distance between Inception features of generated and real images:

$$\text{FID} = \|\mu_g - \mu_r\|_2^2 + \text{tr}(\Sigma_g + \Sigma_r - 2(\Sigma_g \Sigma_r)^{1/2})$$

- 10k EMA samples, Euler ODE, $N=50$
- evaluated every 25 epochs

Convergence: validation loss ↓

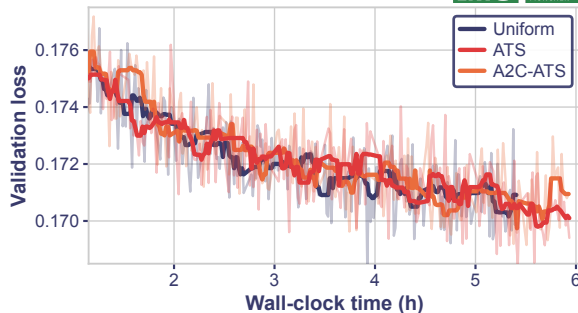
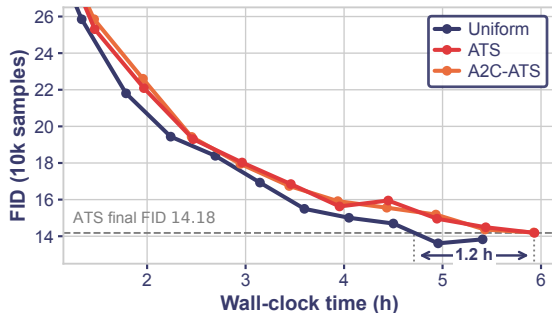
Flow-matching loss on held-out data, always drawn at *uniform* t :

$$\text{Val. loss} = \mathbb{E}_{t \sim \mathcal{U}(0,1)} \|v_\theta(x_t, t) - (x_1 - x_0)\|_2^2$$

- the *training* loss is biased by each run's own t -density
- uniform t makes all samplers comparable

H1.1 – At Equal Compute: Uniform Leads

H1.1: ATS leads to faster convergence than uniform.



Training comparison at equal compute

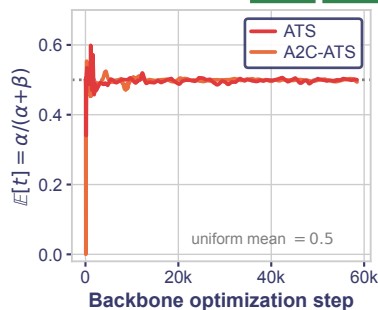
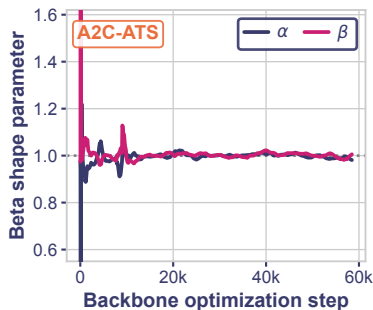
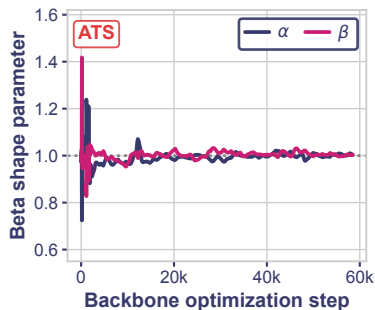
Qualitative Comparison

details

- ATS / A2C-ATS curves shift right by the $\approx 10\%$ overhead ($\approx +0.5$ h)
- **FID:** uniform lies below at every budget and reaches ATS's final value 1.2 h sooner
- **Val. loss:** indistinguishable — the adaptive strategies never lead

H1.1 – Why: The Policy Reverts to Uniform

H1.1: ATS leads to faster convergence than uniform.

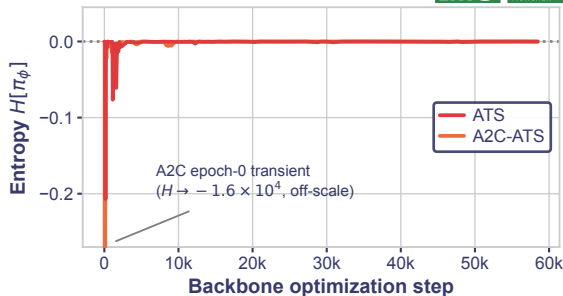
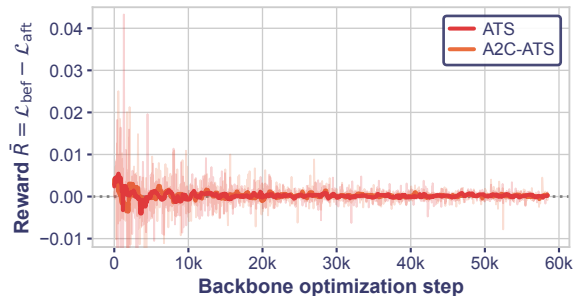


The adaptive policies become uniform after an initial warm-up

- Both oscillate over the first ~ 10 epochs ($\mathbb{E}[t]$ swings 0.34–0.60), then settle at $\alpha \approx \beta \approx 1$
 $\Rightarrow \text{Beta}(1, 1) = \mathcal{U}(0, 1)$
- $\mathbb{E}[t]$ then holds at ≈ 0.5 for the remaining ~ 290 epochs; ATS ends at $\text{Beta}(1.00, 1.00)$

H1.1 – Why: The Reward Vanishes

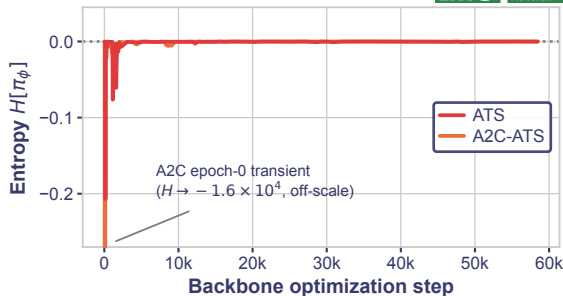
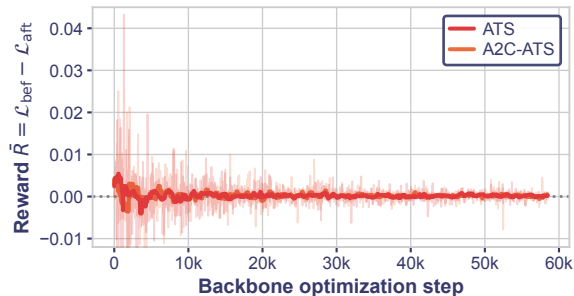
H1.1: ATS leads to faster convergence than uniform.



$\bar{R} \approx 4 \times 10^{-3}$ early, decaying to $\approx 10^{-4}$; $H[\pi_\phi]$ is maximal ($= 0$) exactly at Beta(1, 1) $\Rightarrow$ the entropy term takes over.

H1.1 – Why: The Reward Vanishes

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$\tilde{R} \approx 4 \times 10^{-3}$ early, decaying to $\approx 10^{-4}$; $H[\pi_\phi]$ is maximal (= 0) exactly at Beta(1, 1) $\Rightarrow$ the entropy term takes over.

H1.1 does not hold

No convergence or quality gain over uniform, at a $\approx 10\%$ compute premium

$\Rightarrow$ Loss-reduction reward signal too weak and too static.

E1 – Hypothesis 1.2

H1.1 ATS leads to faster convergence than uniform.

× **refuted**

H1.2 The A2C value baseline lowers the policy-gradient variance.

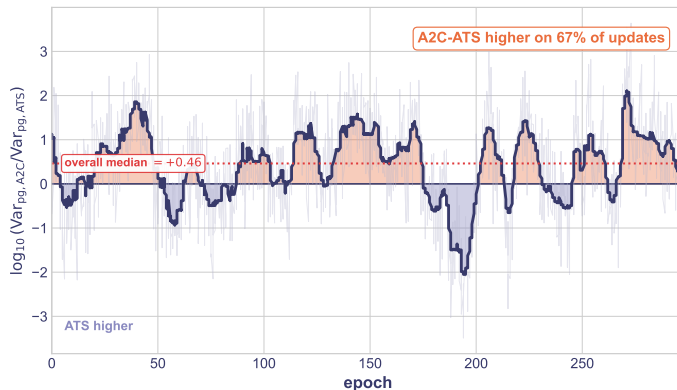
► **now**

H1.3 The curvature (LTE proxy) is a valid reward signal.

H1.4 Adaptive timestep sampling outperforms static non-uniform sampling.

H1.2 – The Critic *Adds* Variance

H1.2: The A2C value baseline lowers the policy-gradient variance.

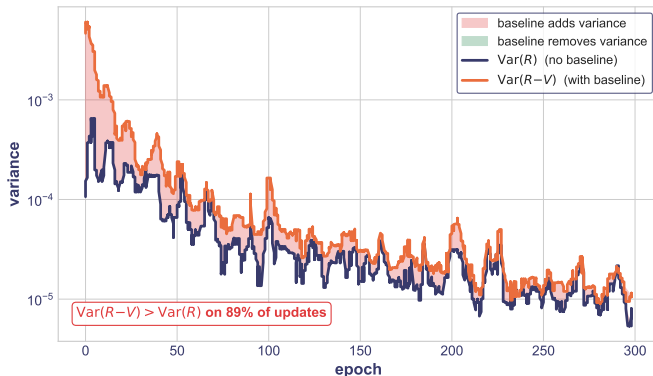


A2C baseline does not reduce the policy-gradient variance; it mildly raises it

- A2C-ATS carries *higher* variance on 67% of updates
⇒ Policy-gradient variance by a median factor of ≈ 2.9 higher (median log-ratio +0.46)

H1.2 – The Critic Injects More Noise Than It Cancels

H1.2: The A2C value baseline lowers the policy-gradient variance.



$$\text{Var}(R - V) = \text{Var}(R) + \text{Var}(V) - 2 \text{Cov}(R, V)$$

	Median	25%	75%
<i>Variances / covariance ($\times 10^{-5}$)</i>			
$\text{Var}(R)$	2.8	1.0	8.6
$\text{Var}(V)$	1.0	0.4	3.4
$\text{Cov}(R, V)$	0.0	-0.2	0.2
Added: $\text{Var}(V) - 2\text{Cov}$	0.9	0.3	3.4
$\text{Var}(R - V)$	4.5	1.8	14.0
$\rho = \text{Cov} / (\sigma_R \sigma_V)$	-0.003	-0.12	0.11

Intuition: every A2C policy update is scaled by the critic's error

- $\hat{g}_{\text{pg,A2C}} = (R - V_\psi) \nabla_{\phi_k} \log \pi_{\phi_k}(t|x_1) \Rightarrow$ variance reduction requires $2 \text{Cov}(R, V) > \text{Var}(V)$
- Value uncorrelated to reward $\Rightarrow$ critic adds variance instead of canceling it

H1.2 – Why the Baseline Fails to Track the Reward

H1.2: The A2C value baseline lowers the policy-gradient variance.

The reward signal is dead

- $|R| \sim 10^{-4}$, $\text{Var}(R) \sim 10^{-5}$ on the straight CondOT interpolant

⇒ almost no per-image reward structure left for a critic to regress on

The reward is non-stationary

- R is the loss reduction of the *backbone*
⇒ moving target

⇒ the critic lags behind and fits outdated network states

H1.2 is refuted: The critic *raises* the policy-gradient variance.

- The value network never reaches the conditional mean reward: $\rho \approx 0$ throughout training
- An uncorrelated baseline cancels nothing ($2 \text{Cov} \approx 0$) and still adds its own $\text{Var}(V)$

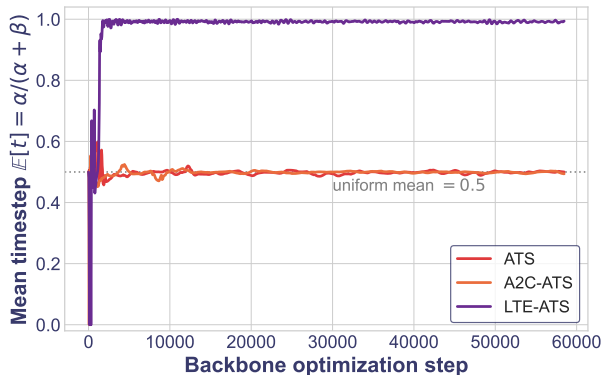
⇒ $\text{Var}(R-V) > \text{Var}(R)$ on 89% of updates

E1 – Hypothesis 1.3

- | | | |
|-------------|---|------------------|
| H1.1 | ATS leads to faster convergence than uniform. | × refuted |
| H1.2 | The A2C value baseline lowers the policy-gradient variance. | × refuted |
| H1.3 | The curvature (LTE proxy) is a valid reward signal. | ► now |
| H1.4 | Adaptive timestep sampling outperforms static non-uniform sampling. | |

H1.3 – The LTE Policy Collapses onto $t \approx 1$

H1.3: The curvature (LTE proxy) is a valid reward signal.

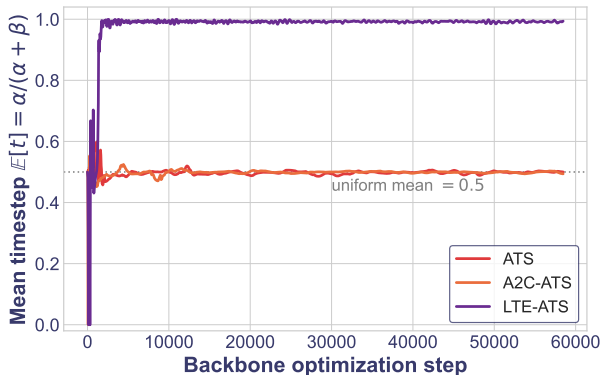


LTE policy gets trapped in data proximal regime

- $\mathbb{E}[t]$ rises $0.5 \rightarrow > 0.9$ within ~ 1400 steps (≈ 7 epochs)
- Policy then freezes at $t \approx 0.994$
- The Beta density is near-degenerate: $\alpha \approx 675, \beta \approx 4.1$

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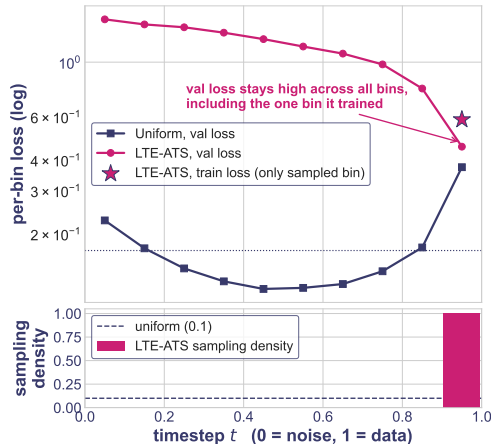
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- The Beta density is near-degenerate: $\alpha \approx 675, \beta \approx 4.1$

H1.3 is refuted — LTE is not a useful reward signal

- Policy responds to the reward strongly, but feedback loop does not close $\Rightarrow$ Action (training at $t \approx 1$) does not influence Reward (curvature)

H1.3 – Why the Collapse Never Corrects Itself

H1.3: The curvature (LTE proxy) is a valid reward signal.



Training <> Inference disconnect

- Training minimizes *conditional* loss and straightens each (x_0, x_1) interpolant
- Inference integrates the *marginal* field
 $\Rightarrow$ straightening conditionals does not straighten that marginal average
- Absorbing mode: Reward is largest at $t \rightarrow 1$, training there never lowers it $\Rightarrow$ policy never leaves.

Result: The backbone underfits heavily

- 9 of 10 t -bins receive zero samples
- Validation loss 1.119 (vs. 0.171)
- FID 395.2 $\Rightarrow$ no longer generative

E1 – Hypothesis 1.4

- | | | |
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H1.4 – Benchmarking Adaptive vs. Static t-Sampling

H1.4: Adaptive timestep sampling outperforms static non-uniform sampling.

Sampler	FID ↓	Val. loss ↓	Wall-clock (h) ↓	NFE _{fwd-eq} ↓
<i>Non-adaptive</i>				
Uniform	13.83	0.1709	5.41	236 910
outlier Logit-normal	13.80	0.1738	5.39	236 910
Bernoulli (data, 0.8)	13.60	0.1697	5.39	236 910
Bernoulli (data, 0.9)	13.84	0.1699	5.40	236 910
Beta (data)	13.70	0.1707	5.41	236 910
Beta (noise)	14.62	0.1704	5.44	236 910
Bernoulli (noise, 0.2)	17.34	0.1704	5.42	236 910
<i>Adaptive</i>				
ATS	14.18	0.1694	5.93	248 910
A2C-ATS	14.21	0.1709	5.93	248 910
LTE-ATS	395.2	1.1194	5.44	239 910

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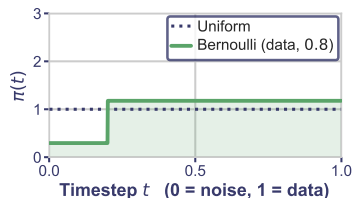
H1.4 is refuted — adaptivity costs more and delivers less

- ATS 14.18, A2C-ATS 14.21 — both *above* uniform's 13.83 in FID
- LTE-ATS collapsed outright (FID 395.2, val. loss 1.12)
- and they pay for it: +12,000 NFE_{fwd-eq}, +0.5 h $\approx$ +10% wall-clock

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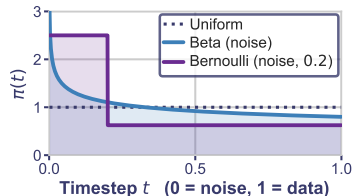
Static non-uniformity is free — and marginally ahead

- every static prior: the same 236 910 NFE_{fwd-eq} and ≈ 5.4 h as uniform
- the gains it buys are tiny: 13.83 $\rightarrow$ 13.60
- best strategy of the suite: Bernoulli (data, 0.8) — a hand-set prior, no policy

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The bias *direction* matters — oversampling the noise boundary is harmful

- Beta: FID = 13.70 → 14.62 (+6.7%)
- Bernoulli: FID = 13.60 → 17.34 (+37.6%)

- | | | |
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Adaptive Timestep Sampling as ported does not transfer to CondOT Flow Matching

- Loss reduction reward is weak ($|R| \sim 10^{-4}$) $\Rightarrow$ admits no adaptive structure to exploit
- LTE reward is *not* weak but mode-peaked and non-decaying $\Rightarrow$ collapses the policy

$\Rightarrow$ Adaptivity never earns back its $\approx 10\%$ overhead.

$\Rightarrow$ Static priors deliver for free what the online policy never found.

E1 left questions open

- Is there an exploitable structure across timesteps? — is it dynamic or static?
- How can timestep sampling exploit this structure?

H2.1 The per-timestep loss and gradient-variance landscapes of CondOT FM are not uniform. **► now**

H2.2 Biasing the sampler toward noise or data reshapes these landscapes.

H2.1 – The Per- t Loss Landscape

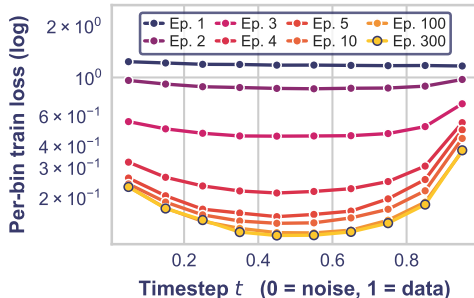
H2.1: The per-timestep loss and gradient-variance landscapes of CondOT FM are not uniform.

$$\mathcal{L}_t = \mathbb{E} \left\| v_\theta(x_t, t) - \underbrace{(x_1 - x_0)}_u \right\|^2 = \underbrace{\mathbb{E} \left\| \mathbb{E}[u | x_t] - v_\theta(x_t, t) \right\|^2}_{\text{reducible by training}} + \underbrace{\text{Cov}(u | x_t)}_{\text{irreducible floor}}$$

Noise end $t \rightarrow 0$

$$\text{Cov}(u | x_t) = \frac{\text{Cov}(x_1 | x_t)}{(1-t)^2}$$

- $x_t \approx x_0$: data unknown
- MSE floor = $\text{Var}(x_1) = \sigma^2 < 1$



Data end $t \rightarrow 1$

$$\text{Cov}(u | x_t) = \frac{\text{Cov}(x_0 | x_t)}{t^2}$$

- $x_t \approx x_1$: noise unknown
- MSE floor = $\text{Var}(x_0) = 1$

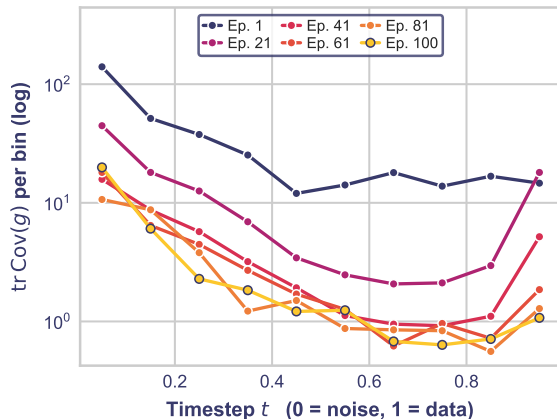
Loss landscape dominated by static, irreducible floor after first epochs

- Epochs 1–2: loss nearly uniform across t (spread $\leq 1.13\times$)
- Epoch 5: reducible part gone $\Rightarrow$ U forms, $\approx 3\times$ spread, taller data arm
- Epochs 10 $\rightarrow$ 300: almost static — a property of the objective, not a moving target

Detailed MSE decomposition

H2.1 – Gradient Variance per t Bin

H2.1: The per-timestep loss and gradient-variance landscapes of CondOT FM are not uniform.



Variance peaks at the boundaries

- noise end: $\text{tr Cov}(g) \approx 13.6$
- interior minimum ($t \approx 0.85$): ≈ 0.6
- data end: ≈ 1.3

$\Rightarrow \approx 22 \times$ noise end vs. min, $\approx 2 \times$ data end vs. min

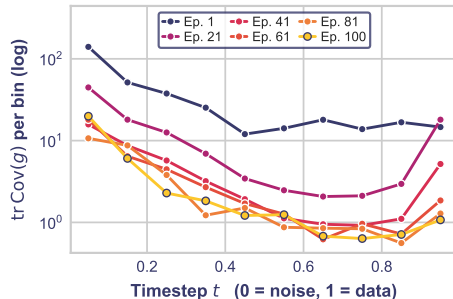
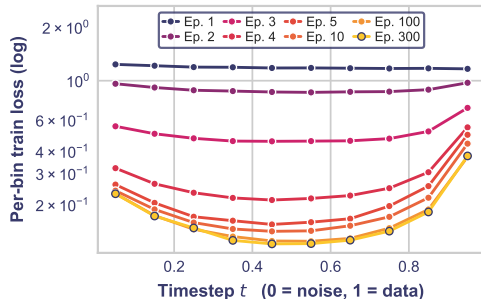
Similar shape throughout training

- The noise end is the maximum (12–36 $\times$ min.)
- Training lowers the level, not the shape.

Estimation of the per-bin gradient variance

H2.1 – Interpreting the Loss and Gradient Landscapes

H2.1: The per-timestep loss and gradient-variance landscapes of CondOT FM are not uniform.



H2.1 holds

- Loss U-shaped ($\approx 3 \times$ variation), static after ~ 10 epochs
- Gradient variance non-uniform too ($\approx 22 \times$), peak at the noise end

Profiles might explain E1 results

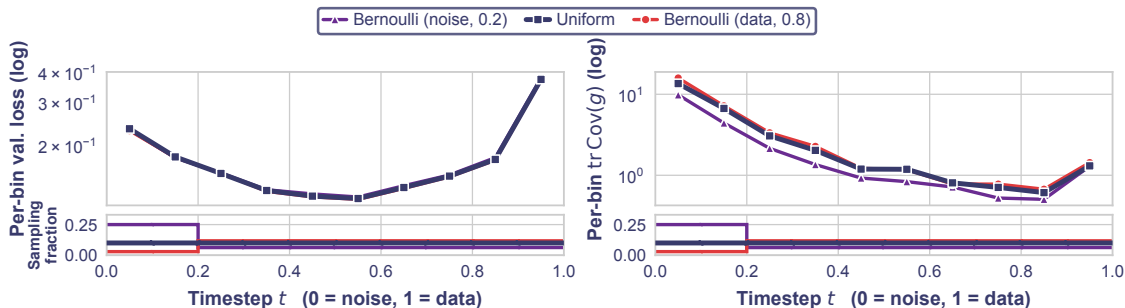
- Static profiles $\Rightarrow$ little for an adaptive policy to exploit
- Noise end hardest, and near-pure noise $\Rightarrow$ oversampling wastes compute
- Data end also hard, but potentially more important

E2 – Hypothesis 2.2

- H2.1 The per-timestep loss and gradient-variance landscapes of CondOT FM are not uniform. ✓ **holds**
- H2.2** Biasing the sampler toward noise or data reshapes these landscapes. ► **now**

H2.2 – t-Distribution Does Not Impact the Landscapes

H2.2: Biasing the sampler toward noise or data reshapes these landscapes.



Instrumented reruns, 100 epochs. Val. loss: mean of the last 20 epochs; tr Cov(g): mean of the last 4 probes (ep. 71, 81, 91, 100).

H2.2 refuted — Final loss invariant...

Same U to within 0.007 per bin:

- Starving a regime does not raise its held-out loss
- Flooding it does not lower it

... & gradient variance keeps shape

- Same noise-dominated U
- Only the *level* shifts (9.8/13.6/15.9 at the noise bin) uniformly in t

H2.1 The per-timestep loss and gradient-variance landscapes of CondOT FM are not uniform.

✓ **holds**

H2.2 Biasing the sampler toward noise or data reshapes these landscapes.

✗ **refuted**

No adaptation needed — static non-uniformity suffices

- Difficulty (loss, gradient variance) is strongly non-uniform in t ($3\times$ / $22\times$)
- ...but almost static: shape set within ~ 10 epochs
- $\Rightarrow$ nothing left for a policy to track

The t distribution acts only through allocation

- It does not shape the difficulty profile — only *where* the compute goes
- Difficult noise boundary wastes compute on gradient noise
 $\Rightarrow$ Possible explanation for the harmful noise bias

Allocation Pricing

Experiment 3 – Impact on Inference Trajectories

The samplers act only during training — does anything survive into generation?

H3

The t -sampling distribution imprints on the inference trajectories.

► now

H3 – Three Geometric Diagnostics

H3: The t -sampling distribution imprints on the inference trajectories.

Path deviation

$$\text{PD} = 1 - \cos(v_\theta(x_t, t), x_t - x_0)$$

Angle between the current velocity and the path travelled so far.

- 0 iff the path is a straight ray
- Invariant to velocity magnitude

Kinetic transport energy

$$W_{\text{kin}} = \int_0^1 \mathbb{E} \|v_\theta(x_t, t)\|_2^2 dt$$

Integral over the squared expected velocity norm.

- Curved or non-constant-speed paths need more energy

Local truncation error

$$\widehat{\text{LTE}} = \left\| \frac{\Delta t}{2} (v_\theta(\tilde{x}_{t+\Delta t}) - v_\theta(x_t)) \right\|_2^2$$

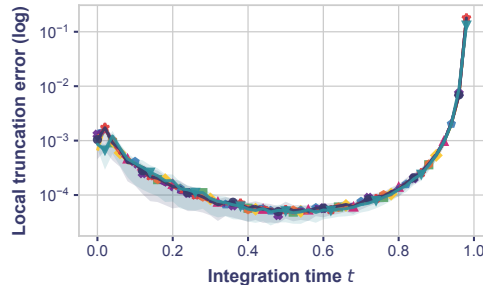
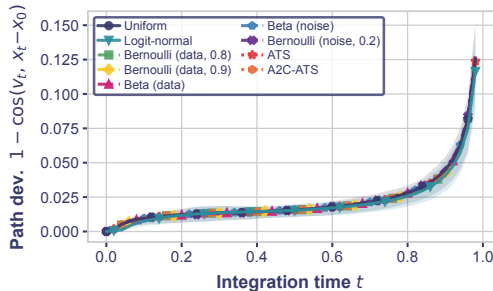
The Euler–Heun discrepancy as proxy for integration error.

Methodology

- All diagnostics computed *post-training* on the epoch-300 models
- Integrate $\dot{x}_t = v_\theta(x_t, t)$ from $t=0$ to 1, Euler, $N=50$
- 256 noise draws *shared* across all strategies
- Each metric reported as its mean $\pm$ the standard error of that mean

H3 – One Shared Inference Geometry

H3: The t -sampling distribution imprints on the inference trajectories.



H3 refuted – identical geometry

details

Even though the launch-regime density $p(t \rightarrow 0)$ spans $\approx 18\times$ across the fixed priors (0.14 to 2.5; logit-normal ≈ 0)
 $\Rightarrow$ inference trajectories share identical profile

Stable also with $N < 50$

Logit-normal faster and straighter

Near-straight interior, late turn

- deviation ≈ 0.02 in the interior, ≈ 0.12 only as $t \rightarrow 1$
- LTE is a U in t : minimal near $t \approx 0.45$, spiking at $t \rightarrow 1$
- the late turn commits the field to a mode: *stationary* under training, and the reward LTE-ATS chased (H1.3)

Outline

- 1 Introduction to Flow Matching
- 2 Related Work on Training Efficiency
- 3 Adaptive Timestep Sampling for Flow Matching
- 4 Experiments
- 5 Conclusion**

1 Adaptive sampling gives no benefit over uniform (E1).

Val. loss 0.1694 vs. 0.1709, FID 14.18 vs. 13.83, at +5.1% NFE_{fwd-eq} and +9.6% wall-clock. Policy explores for ten epochs ($\mathbb{E}[t]$ swings 0.34–0.60), then reverts to Beta(1, 1) once the reward decays entropy takes over.

2 Static noise-bias is harmful (E1).

FID 13.83 $\rightarrow$ 14.62 and 17.34; The data-proximal direction gives no reliable gain: 13.60, 13.80, 13.83 sit in a 0.23-point band inside run-to-run variation.

3 The A2C critic raises policy-gradient variance (E1).

$\rho(R, V) \approx 0$, so the baseline adds uncorrelated noise; higher on 67% of updates. The likely cause is the weak reward.

4 The curvature reward collapses the policy (E1, E3).

$t \rightarrow 0.994$ within ~ 1400 steps, val. loss 1.119, FID 395.2: minimizing *conditional* loss does not straighten the *marginal* field.

5 The difficulty profile is structured but mostly stationary (E2).

Loss U (max at data), gradient-variance U spanning $\approx 20\times$ (max at noise), both fixed within a few epochs. A sampler reallocates compute — it does not alter the learning task.

6 Training biases leave the flow geometry (mostly) unchanged (E3).

Curvature, W_{kin} and LTE profile identical (exception: logit-normal), unchanged at every integration budget.

In CondOT FM the **non-stationary signal** an adaptive policy could exploit is **short-lived** and close to absent after a few epochs $\Rightarrow$ A **static difficulty structure** persists that a **fixed prior** exploits more cheaply.

Findings

No benefit over uniform (E1)

FID 14.18 vs. 13.83 at +5.1% NFE

Static noise-bias is harmful (E1)

FID 13.83 $\rightarrow$ 17.34; data-proximal gain within run-to-run noise

The A2C critic raises gradient variance (E1)

$\rho(R, V) \approx 0$: the baseline adds uncorrelated noise

The curvature reward collapses the policy (E1, E3)

$t \rightarrow 0.994$, FID 395.2: conditional $\neq$ marginal

Difficulty is structured but stationary (E2)

loss U, $\approx 20 \times$ gradient-variance U, both fixed early

Flow geometry (mostly) unchanged (E3)

curvature, W_{kin} , LTE identical at every budget

Limitations

One seed, one small setting

single model suite on CIFAR-10 32×32 , one 35.7M U-Net

Quality rests on FID alone

two ImageNet feature moments, biased at 10k

Overhead only approximated

NFE_{fwd-eq} counts units, not FLOPs; wall-clock hardware dependent

Most metrics are estimated

high estimation error; summary statistics hide variation

Hyperparameters carried over

from the DDPM ϵ -setting, not re-ablated

$\Rightarrow$ Multi-seed, larger scale

latent FM, DiT, ImageNet: richer per- t dynamics may revive a non-stationary reward

$\Rightarrow$ Add DINOv2-FD, CMMD, precision/recall

decides whether the data-bias edge is real

$\Rightarrow$ Scalability study

policy cost is backbone-independent — its budget share shrinks with model size

Map the reward along t

read a sampler off the E2 gradient-variance curve — non-stationary only early

Isolate the failing factor

VP path, loss weighting, ϵ vs. velocity; match the optimizer to a weak reward (PPO, entropy schedule)

Future work

Conclusion and implications

The premise of ATS transfers to Flow Matching; the mechanism does not. Straight CondOT paths leave a weak, decaying reward with no stable non-uniform optimum — adaptation pays estimation overhead for structure a fixed prior already captures.

For practice: avoid the noise boundary, take the free static prior.

For future schemes: the exploitable axis seems to be the *gradient variance* — invisible to a loss-reduction reward.

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- Boundary coupling optimization
- The three dials of the objective
- REINFORCE: the proofs
- A2C: policy-gradient variance
- A2C: the baseline condition

E1 – Efficiency

- Loss vs. wall-clock
- H1.1 – qualitative comparison
- H1.4 – logit-normal outlier
- Train vs. val. loss

E2 – Learning process

- Per-timestep statistics
- Why the loss is U-shaped
- Gradient-probe estimator
- H2.2 – gradient SNR prices the allocation

E3 – Inference

- Training densities
- Kinetic power
- H3 – logit-normal outlier
- H3 – few-step regime
- FID vs. step budget

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Push-forward and change of variables

The flow transports the whole prior, and the evolving density follows from the change-of-variables formula:

$$p_t = [\phi_t]_* p_0, \quad p_t(x_t) = p_0(\phi_t^{-1}(x_t)) \det\left(\frac{\partial \phi_t^{-1}(x_t)}{\partial x_t}\right)$$

The Jacobian determinant accounts for local expansion or compression, enforcing exact mass conservation.

The continuity equation

A pair (p_t, v_t) is valid iff it conserves mass instantaneously:

$$\frac{\partial}{\partial t} p_t(x_t) + \nabla \cdot (p_t(x_t) v_t(x_t, t)) = 0$$

Deriving a u_t that satisfies this for a complex $q(x_1)$ is intractable — which is exactly why Flow Matching builds *conditional* fields that satisfy it by construction.

Existence and uniqueness

Neural vector fields are Lipschitz continuous, so Picard–Lindelöf guarantees a unique solution: trajectories never intersect.

One family: Gaussian conditional paths

$$p_t(x_t|x_1) = \mathcal{N}(x_t; \alpha_t x_1, \beta_t^2 I), \quad u_t(x_t|x_1) = \left(\dot{\alpha}_t - \frac{\dot{\beta}_t}{\beta_t} \alpha_t \right) x_1 + \frac{\dot{\beta}_t}{\beta_t} x_t$$

Any differentiable, monotone (α_t, β_t) with $\alpha_0 = \beta_1 = 0$ and $\alpha_1 = \beta_0 = 1$ is admissible: the conditional field above is the unique one generating that path, so **choosing** (α_t, β_t) **chooses the regression target**.

CondOT

$\alpha_t = t, \beta_t = 1 - t \Rightarrow u_t = x_1 - x_0$,
constant speed, straight conditional lines,
minimal conditional kinetic energy;
 $\text{SNR}(t) = \left(\frac{t}{1-t}\right)^2$. The FM /
rectified-flow standard [18].

VP (diffusion)

$\alpha_t^2 + \beta_t^2 = 1$ (e.g. $\sin / \cos$) $\Rightarrow$ the
marginal variance is preserved at every t .
This is the DDPM / score-based path [8]:
strongly curved near both ends, hence
many NFE.

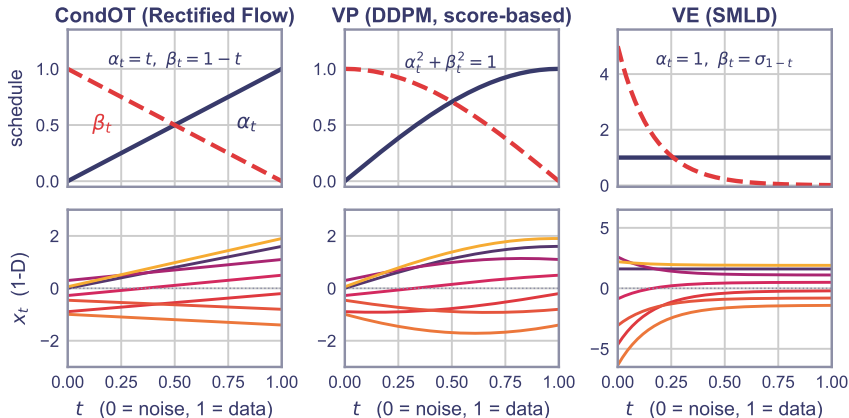
VE (score matching)

$\alpha_t \equiv 1, \beta_t = \sigma_{1-t}$ geometric with
 $\sigma_{\max} \gg$ data scale [27]. The prior is only
reached approximately, and the huge
dynamic range makes the path the most
curved of the three.

Why this matters for the thesis

In diffusion the path is *fixed* by the forward SDE and entangled with the loss weighting; in Flow Matching it is a free choice, decoupled from the objective. Committing to CondOT locks the geometry — and leaves the sampling distribution $\pi(t)$ as the one remaining lever.

Appendix – CondOT, VP and VE Side by Side



Reading

Top: the schedules (α_t scales the data, β_t the noise). *Bottom:* seven conditional interpolants $x_t = \alpha_t x_1 + \beta_t x_0$ in 1-D, one colour per (x_0, x_1) pair. Only CondOT moves at constant speed along a straight line; VP bends because $\alpha_t^2 + \beta_t^2 = 1$ forces a trigonometric turn; VE ($\sigma_{\max}=5$) collapses nearly all its motion into $t < 0.25$.

Theorem [17]

For any valid conditional field $u_t(x_t|x_1)$, the two objectives differ by a constant in θ , hence $\nabla_{\theta} \mathcal{L}_{\text{FM}}(\theta) = \nabla_{\theta} \mathcal{L}_{\text{CFM}}(\theta)$: a minibatch of CFM is an **unbiased** estimator of the intractable marginal gradient.

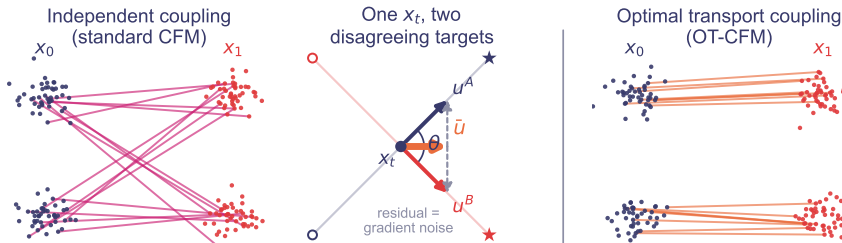
Proof in three steps

- 1 **Expand the square.** $\mathcal{L}_{\text{FM}} = \mathbb{E}[\|v_t\|^2 - 2\langle v_t, u_t(x_t) \rangle] + C_1$, with $C_1 = \mathbb{E}\|u_t(x_t)\|^2$ independent of θ .
- 2 **Rewrite both terms over the joint.** Since $p_t(x_t) = \int p_t(x_t|x_1)q(x_1)dx_1$, the quadratic term is unchanged under $x_1 \sim q$, $x_t \sim p_t(\cdot|x_1)$. For the cross-term, substitute the definition of the marginal field and exchange the integrals:

$$\mathbb{E}_{x_t \sim p_t} \langle v_t, u_t(x_t) \rangle = \int \int \int \langle v_t, u_t(x_t|x_1) \rangle p_t(x_t|x_1)q(x_1) dx_1 dx_t dt = \mathbb{E}_{x_1, x_t} \langle v_t, u_t(x_t|x_1) \rangle$$

- 3 **Complete the square.** Adding and subtracting $\mathbb{E}\|u_t(x_t|x_1)\|^2$ gives $\mathcal{L}_{\text{FM}} = \mathcal{L}_{\text{CFM}} + C$. $\square$

Step 2 is the crucial one: it replaces an expectation over the *intractable* marginal target by one over the *tractable* per-sample target. Proof after Holderrieth and Erives [9].



Methods for Boundary Coupling Optimization

- **Mini-batch OT** [28, 23]: pair to minimize $W_2 \Rightarrow$ non-crossing paths
- **Immiscible Diffusion** [16]: L_2 assignment, up to $3 \times$ faster
- **Augmented FM** [33]: auxiliary variables, $< 0.5\%$ overhead
- **Stable Target Field** [32]: average u over a reference batch

◀ back to talk

Three dials in one objective — and only one is still free

$$\mathcal{L}_{\text{CFM}}(\theta) = \mathbb{E}_{t \sim \pi(t), x_1 \sim q, x_0 \sim \mathcal{N}(0, I)} \left[w(t) \left\| v_t(\alpha_t x_1 + \beta_t x_0; \theta) - u_t(\cdot | x_1) \right\|^2 \right]$$

- **Noise schedule** (α_t, β_t) — sets the SNR at each t

Locked in FM: CondOT fixes $\text{SNR}(t) = \left(\frac{t}{1-t}\right)^2$; changing it re-curves the path [18]

- **Loss weighting** $w(t)$ — rescales gradients

CondOT regresses a finite-variance target, so $w(t) \equiv 1$ [4]; re-weighting [2, 7] also *pays full compute* for timesteps it then down-scales

- **Timestep sampling** $\pi(t)$ — changes the *batch composition*

Reallocates real SGD updates and FLOPs [34, 12]

Consequence

- In DDPM all three were entangled through the variational bound [11]
- Flow Matching decouples them — and locks two of the three

⇒ $\pi(t)$ is the remaining dial

◀ back to talk

Appendix – REINFORCE: The Proofs

Policy $\pi_\phi(a|s)$, reward R with $\nabla_\phi R = 0$, eligibility $\mathbf{e} = \nabla_\phi \log \pi_\phi$. In ATS: $a = t, s = x_1, R = \tilde{\Delta}_k^t$.

1. Log-derivative trick $\Rightarrow$ policy gradient

$$\begin{aligned}\nabla_\phi J(\phi) &= \nabla_\phi \int \pi_\phi(a|s) R \, da && \text{(expectation as integral)} \\ &= \int \nabla_\phi \pi_\phi(a|s) R \, da && \text{(Leibniz rule)} \\ &= \int \pi_\phi(a|s) \nabla_\phi \log \pi_\phi(a|s) R \, da && (\nabla \pi = \pi \nabla \log \pi) \\ &= \mathbb{E}_{a \sim \pi_\phi} [R \nabla_\phi \log \pi_\phi(a|s)] && \text{(back to expectation)}\end{aligned}$$

- Swapping ∇_ϕ and $\int$ needs π_ϕ differentiable in ϕ with dominated derivative – satisfied by $\text{Beta}(\alpha_\phi, \beta_\phi)$ on $(0, 1)$
- $\nabla_\phi R = 0$ is what lets R pass through ∇_ϕ : in ATS the reward is produced by the *backbone*, whose θ are independent of ϕ

2. Unbiasedness and the baseline

variance

Update $\Delta\phi = \alpha(R - b)\mathbf{e}$, with b independent of a given s :

$$\begin{aligned}\mathbb{E}[b \nabla_\phi \log \pi_\phi(a|s)] &= b \int \pi_\phi(a|s) \nabla_\phi \log \pi_\phi(a|s) \, da \\ &= b \nabla_\phi \underbrace{\int \pi_\phi(a|s) \, da}_{=1} = b \nabla_\phi(1) = 0, \\ \Rightarrow \mathbb{E}[\Delta\phi | \phi] &= \alpha \mathbb{E}[R \mathbf{e}] - \underbrace{\alpha \mathbb{E}[b \mathbf{e}]}_{=0} = \alpha \nabla_\phi \mathbb{E}[R | \phi]. \quad \square\end{aligned}$$

- Holds for *every* action-independent b : $b = 0$ (ATS) and $b = V_\psi(x_1)$ (A2C) share one expectation
- Says nothing about variance – set by $\text{Var}(R - b)$

Proof after Williams [30]; entropy bonus from Williams and Peng [31], as in A3C [20].

◀ back to talk

Appendix – A2C: Why Only the Second Moment Differs

Notation: $R = \tilde{\Delta}_k^t$, $V = V_\psi(x_1)$, score $S_k(t) = \nabla_{\phi_k} \log \pi_{\phi_k}(t|x_1)$, estimators $\hat{g}_{\text{ATS}} = R S_k$ and $\hat{g}_{\text{A2C}} = (R - V)S_k$.



1. Why the *trace* of the covariance

$\hat{g} \in \mathbb{R}^{|\phi|}$, so its second-order behaviour is a matrix — and matrices are not ordered. The trace turns it into the one scalar that matters for optimization,

$$\text{Tr Cov}(\hat{g}) = \mathbb{E} \|\hat{g} - \mathbb{E}\hat{g}\|^2 = \mathbb{E} [\|\hat{g}\|^2] - \|\mathbb{E}\hat{g}\|^2,$$

- the expected squared *deviation* of one stochastic update — the noise constant σ^2 of SGD convergence bounds [5];
- invariant under orthogonal reparametrizations of ϕ ;
- *linear*, so it survives the law of total covariance (unlike $\det$ or $\|\cdot\|_{\text{op}}$).

2. Split, then cancel everything but one term

Law of total covariance, on the image:

$$\text{Tr Cov}(\hat{g}) = \underbrace{\mathbb{E}_{x_1} [\text{Tr Cov}_{t,x_0}(\hat{g}|x_1)]}_{\text{stochastic}} + \underbrace{\text{Tr Cov}_{x_1}(\mathbb{E}_{t,x_0}[\hat{g}|x_1])}_{\text{batch}}$$

$\mathbb{E}_t[VS_k] = 0$ equalizes the conditional means, so the *batch* term is identical. Inside the stochastic term,

$$\text{Tr Cov}_{t,x_0}(\hat{g}_\bullet|x_1) = \mathbb{E}[A_\bullet^2 \|S_k\|^2 | x_1] - \|\mathbb{E}[\hat{g}_{\text{ATS}}|x_1]\|^2$$

with $A_{\text{ATS}} = R$, $A_{\text{A2C}} = R - V$. The subtracted term is shared $\Rightarrow$ **the estimators differ only in the uncentered second moment** $\mathbb{E}[A_\bullet^2 \|S_k\|^2]$ [29], i.e. exactly the difference on the main slide:

$$\text{Tr Cov}(\hat{g}_{\text{ATS}}) - \text{Tr Cov}(\hat{g}_{\text{A2C}}) = \mathbb{E}[R^2 \|S_k\|^2] - \mathbb{E}[(R - V)^2 \|S_k\|^2]$$

3. The exact (score-weighted) condition

V is constant given x_1 , so the gain of A2C over ATS is

$$\begin{aligned} \mathbb{E}[R^2 \|S_k\|^2 | x_1] - \mathbb{E}[(R - V)^2 \|S_k\|^2 | x_1] \\ = V \left(2 \mathbb{E}[R \|S_k\|^2 | x_1] - V \mathbb{E}[\|S_k\|^2 | x_1] \right) \end{aligned}$$

which is positive iff V lies strictly between 0 and twice the optimal baseline [29]

$$b^*(x_1) = \frac{\mathbb{E}[R \|S_k\|^2 | x_1]}{\mathbb{E}[\|S_k\|^2 | x_1]}.$$

The critic target $V \approx \mathbb{E}_{t, x_0}[R | x_1]$ is the $\|S_k\|^2$ -unweighted approximation of b^* ; the two coincide when the reward is uncorrelated with the score norm.

4. The scalar condition measured in E1

Decoupling $\|S_k\|^2$ leaves a pure reward statement. With $\mathbb{E}[R - V] \approx 0$ for a calibrated critic,

$$\mathbb{E}[(R - V)^2] = \underbrace{(\mathbb{E}[R] - \mathbb{E}[V])^2}_{\text{bias of the critic}} + \text{Var}(R - V)$$

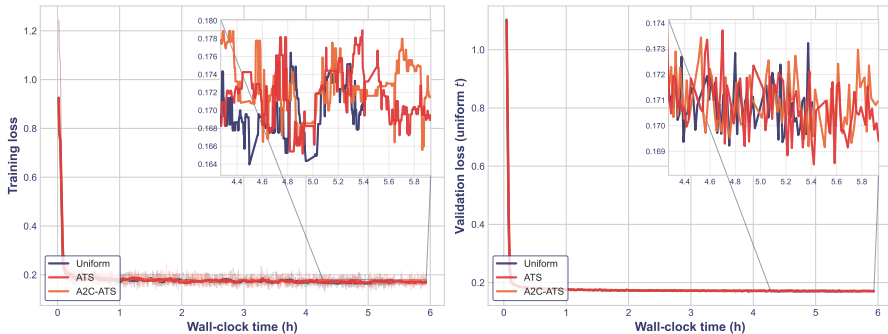
$$\text{Var}(R - V) = \text{Var}(R) + \underbrace{\text{Var}(V)}_{\text{noise injected}} - \underbrace{2 \text{Cov}(R, V)}_{\text{noise explained}}$$

- helps iff $2 \text{Cov}(R, V) > \text{Var}(V) \iff \rho > \sigma_V / (2\sigma_R)$;
- $\rho = 0$: pure surcharge, $\text{Var}(R - V) = \text{Var}(R) + \text{Var}(V)$;
- best case, at the optimal scale $\sigma_V = \rho \sigma_R$:
 $\text{Var}(R - V) = (1 - \rho^2) \text{Var}(R)$ – a critic can only remove the share of the fluctuation it actually predicts.

E1 measures $\rho \approx -0.003$ at $\sigma_V / \sigma_R \approx 0.6$ (needed: $\rho > 0.3$).

◀ back to talk

Appendix – Loss Curves at Equal Wall-Clock

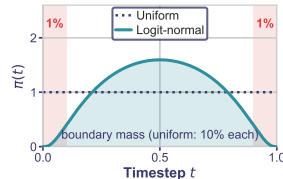
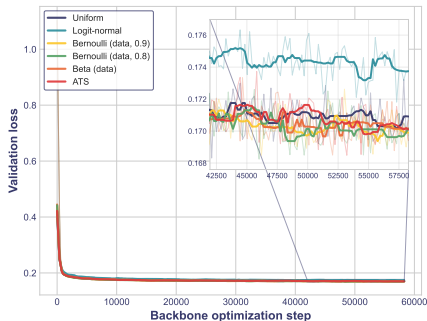
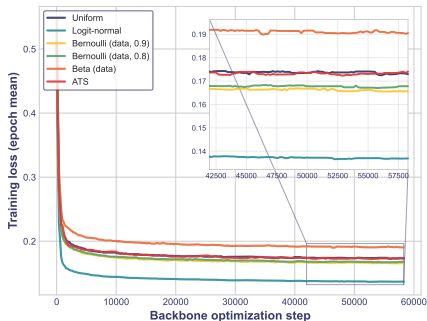


Reading

Left: training loss (logged every 40 steps, raw faint, 25-point rolling median bold); its high-frequency scatter is minibatch estimation error. *Right:* the sampler-independent validation loss per epoch. The adaptive curves are shifted right by their $\approx 10\%$ overhead and never lead uniform.

H1.4 – Train vs. Val. Loss: Logit-Normal Is the Outlier

H1.4: Adaptive timestep sampling outperforms static non-uniform sampling.

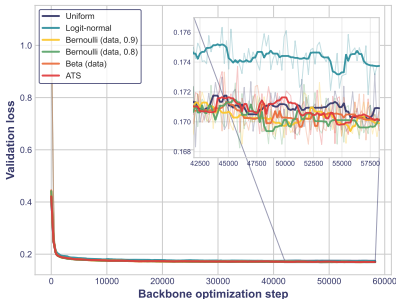
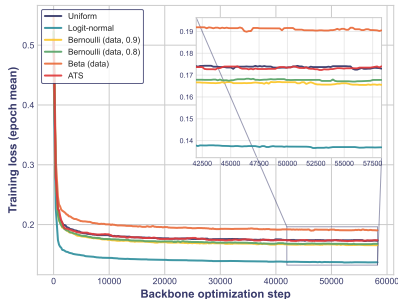


Cause: the starved logit-normal boundaries

details

- train loss = the shared U-shaped landscape weighted by each sampler's *own* t -density
- logit-normal puts $\approx 1\%$ of its draws in each boundary regime $\rightarrow$ reads low on the cheap mid-range
- those boundaries stay underfit, and the val. loss at uniform t charges exactly that

Appendix – Why the Training Loss Cannot Rank Samplers



The two panels disagree

Logit-normal posts the *lowest* training loss (0.137) and the *highest* validation loss (0.174); the rest collapse onto ≈ 0.170 .

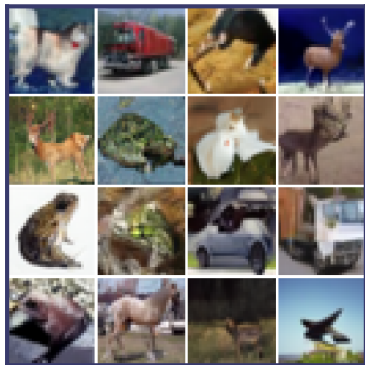
Because t is drawn differently

The epoch loss is the shared U-shaped landscape weighted by each sampler's own density; logit-normal concentrates on the cheap mid-range, so it reads low without being better.

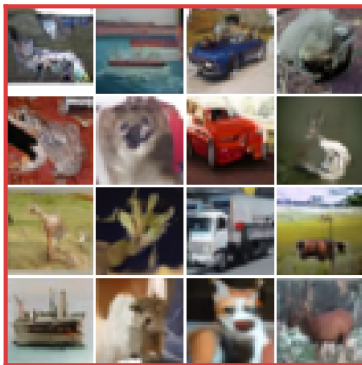
H1.1 – No Visible Difference in the Samples

H1.1: ATS leads to faster convergence than uniform.

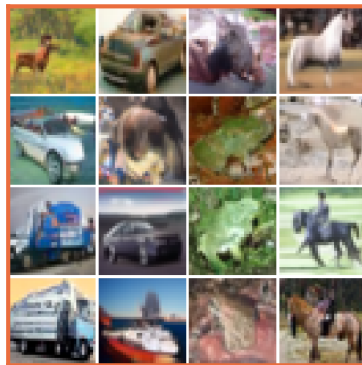
Uniform



ATS



A2C-ATS



Final EMA model of each strategy

Euler, $N=50$; each panel drawn from its own run's noise. Perceptually interchangeable, matching the <1 FID spread.

Regime	t	Loss $\mathcal{L}_t$	tr Cov(g)	$\ \bar{g}\ ^2$	SNR $\ \bar{g}\ ^2 / \text{tr Cov}(g)$
Noise boundary	0.05	0.231	13.56	0.236	0.017
Loss minimum	0.55	0.120	1.19	0.128	0.108
Gradient-variance minimum	0.85	0.181	0.61	0.152	0.249
Data boundary	0.95	0.369	1.31	0.152	0.116

Spread across t

Loss $3.1\times$, gradient variance $\approx 20\times$, but the mean-gradient norm less than $2\times$ — the imbalance is in the *noise* of the gradient, not its direction.

Averaging

Loss over the last 20 epochs, gradient quantities over the last 4 probes (a single $K=8$ probe carries $\approx 50\%$ relative error).

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Reducible and irreducible part

$$\mathcal{L}_t(\theta) = \underbrace{\mathbb{E} \|\mathbb{E}[u \mid x_t, t] - v_\theta(x_t, t)\|^2}_{\text{reducible by training}} + \underbrace{\mathbb{E} \|u - \mathbb{E}[u \mid x_t, t]\|^2}_{\text{irreducible}}$$

Conditioning on x_t ties both endpoints

$$\text{Cov}(u \mid x_t) = \frac{\text{Cov}(x_1 \mid x_t)}{(1-t)^2} = \frac{\text{Cov}(x_0 \mid x_t)}{t^2}$$

- $t \rightarrow 1$: noise enters x_t with weight $(1-t) \rightarrow 0 \Rightarrow$ loss rises toward $\text{Var}(x_0) = 1$.
- $t \rightarrow 0$: data enters with weight $t \rightarrow 0 \Rightarrow$ loss rises toward $\text{Var}(x_1) = \sigma^2 < 1$.

Consequence

Both ends expose a full marginal variance, but of *different* variables — which makes the data boundary the taller arm. Training exhausts the reducible part within a few epochs; the rest is an irreducible floor.

Micro-batch probe

A backward pass returns only a *summed* minibatch gradient, so every tenth epoch each bin b is probed with $K=8$ micro-batches of $m=64$ images, each with fresh noise and a timestep drawn uniformly *within* the bin. With $\bar{g}_b^{(k)}$ the micro-batch gradients and $\hat{g}_b$ their mean:

$$\underbrace{\widehat{\text{tr Cov}}(g)_b = \frac{m}{K-1} \sum_{k=1}^K \|\bar{g}_b^{(k)} - \hat{g}_b\|^2}_{\text{per-bin gradient variance}}, \quad \underbrace{\|\bar{g}_b\|^2 = \|\hat{g}_b\|^2 - \frac{\widehat{\text{tr Cov}}(g)_b}{Km}}_{\text{per-bin gradient mean norm}}, \quad \text{SNR}_b = \frac{\|\bar{g}_b\|^2}{\widehat{\text{tr Cov}}(g)_b}$$

Why debias the norm

$\mathbb{E}\|\hat{g}_b\|^2 = \|\bar{g}_b\|^2 + \text{tr Cov}(g)_b/(Km)$: a finite sample inflates the squared norm, so the variance term is subtracted back out [19, App. A.1].

Unbiased, but noisy

Unbiased for any $K \geq 2$. A variance from K draws has relative error $\sqrt{2/(K-1)} \approx 53\%$, which m cannot reduce $\Rightarrow$ average the last four probes, read orderings.

Sampler-independent

The probe draws its own timesteps, so precision does not depend on how rarely the training sampler visits a bin. Runs outside the optimizer, before clipping, $\approx 1\%$ overhead.

References: estimator and debiasing [19]; per-timestep gradient variance as the difficulty measure [12]; iterations to converge grow with gradient variance [5].

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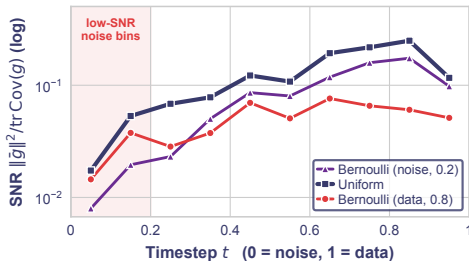
H2.2 – The Gradient SNR Prices the Allocation

H2.2: Biasing the sampler toward noise or data reshapes these landscapes.

$$\text{SNR}_{\text{grad},b} = \frac{\|\widehat{\bar{g}}_b\|^2}{\widehat{\text{tr Cov}}(g)_b} = \frac{\text{signal: strength of the learning direction}}{\text{noise: disagreement between samples}}$$

estimated per t bin from $K=8$ micro-batch probes of $m=64$ images

Estimation details



Same profile under all three priors — the SNR landscape is invariant to the t density.

Low SNR in the noise regime

- Signal *not* weaker, but disagreement $\approx 22\times$ larger
- Mass here mainly buys noise for the *shared* update

Possible explanation for harmful noise bias

Compute spent on two low-SNR noise bins ($t < 0.2$):

- Uniform: 20% $\Rightarrow$ FID 13.83
- Bernoulli (noise, 0.2): 50% $\Rightarrow$ FID 17.34

$\Rightarrow$ Avoiding low-SNR frees compute for more useful updates

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Appendix – How Far Apart the Training Densities Are

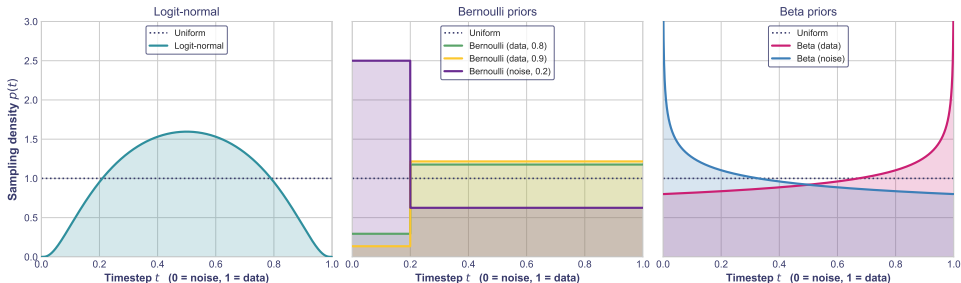
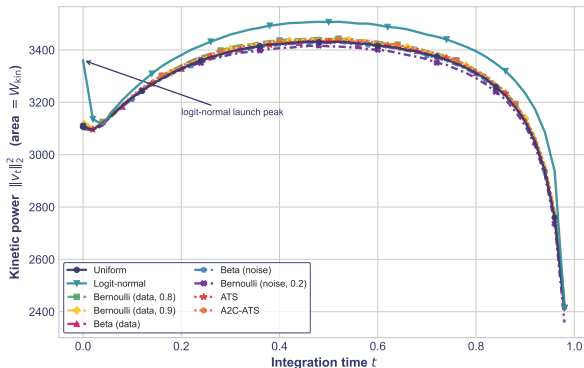


Figure: Density at the launch regime $t \rightarrow 0$. Every trajectory starts on the noise manifold, so the density at $t \rightarrow 0$ measures how much training mass a prior withdraws from it: 0.29 for Bernoulli (data, 0.8), 0.14 for (data, 0.9), 2.5 for Bernoulli (noise, 0.2), and almost zero for logit-normal — the only strategy starving *both* ends, and the only one whose geometry separates.

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Shape

The field starts near the minimum-norm launch value $\|x_0\|_2^2 \approx 3072$, accelerates to a mid-schedule maximum, and decelerates onto the data manifold. Its magnitude measures how much the per-sample targets agree.

The one resolvable trace

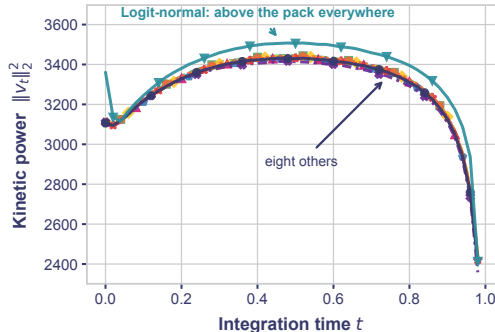
Logit-normal alone sits above the pack (3364 ± 21 vs. 3290 ± 19): its starved noise boundary is underfit, so the launch velocity has not yet settled to $-x_0$. An artifact, not a geometric imprint.

H3 – The One Outlier: Logit-Normal

H3: The t -sampling distribution imprints on the inference trajectories.

Sampler	Path dev. ($\times 10^{-3}$)	W_{kin}	LTE ($\times 10^{-3}$)
Uniform	22.26 ± 0.44	3290 ± 19	3.97 ± 0.038
Logit-normal	20.71 ± 0.42	3364 ± 21	3.07 ± 0.056
Bernoulli (data, 0.8)	21.73 ± 0.44	3302 ± 19	4.00 ± 0.036
Bernoulli (data, 0.9)	21.55 ± 0.43	3299 ± 19	3.99 ± 0.036
Beta (data)	22.09 ± 0.43	3289 ± 19	3.96 ± 0.034
Beta (noise)	22.29 ± 0.43	3286 ± 19	3.94 ± 0.037
Bernoulli (noise, 0.2)	22.52 ± 0.44	3276 ± 18	4.01 ± 0.041
ATS	21.94 ± 0.44	3293 ± 19	3.98 ± 0.037
A2C-ATS	21.95 ± 0.44	3301 ± 19	3.96 ± 0.036
LTE-ATS [†]	50.22 ± 0.10	1341 ± 2	1.56 ± 0.004

Euler, $N=50$, 256 paired draws; $\pm$ is the standard error of the mean (per-trajectory scatter $16\times$ larger). **Bold: the one row that separates.** [†] LTE-ATS is non-generative.



Fastest *and* straightest

static t -sampling densities

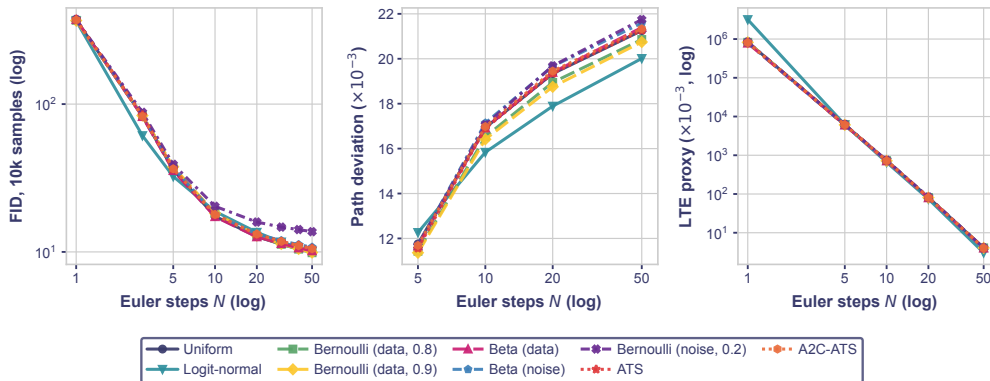
- Logits-normal: starved noise boundary is *underfit* $\Rightarrow$ launch velocity not yet settled to $\mathbb{E}[x_1 - x_0 \mid x_0] = -x_0$
- Interior trained most: lowest path deviation and LTE

$\Rightarrow$ Completely starving *both* ends $\pi(t) \rightarrow 0$ can imprint on trajectories

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H3 – No Hidden Gap in the Few-Step Regime

H3: The t -sampling distribution imprints on the inference trajectories.



Nine converged strategies, one curve at every budget

⇒ Few-step quality governed by the discretization error of equally straight flows — not by the training sampler.

FID table

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Appendix – FID vs. the Step Budget

Sampler	$N=1$	5	10	20	50
Uniform	371.1	35.9	17.4	12.7	10.0
Logit-normal	365.8	32.4	18.8	13.6	10.1
Bernoulli (data, 0.8)	368.7	37.1	18.2	12.8	9.9
Bernoulli (data, 0.9)	367.4	37.7	18.3	12.9	9.9
Beta (data)	373.1	35.5	17.3	12.6	10.2
Beta (noise)	371.9	37.1	18.0	13.2	10.8
Bernoulli (noise, 0.2)	373.2	39.1	20.4	15.9	13.7
ATS	372.5	36.2	17.8	13.1	10.6
A2C-ATS	370.9	36.5	17.9	13.2	10.5
LTE-ATS [†]	404.7	398.0	397.2	395.4	395.1

Nine strategies, one curve

FID on 10k samples, Euler. Every converged strategy inflates by $\approx 3.5\times$ its $N=50$ score at $N=5$; only the two known outliers move. [†]LTE-ATS is non-generative at every budget.

The geometry agrees

Recomputed on the epoch-300 models, noise paired across budgets: the truncation error grows as $\approx N^{-2}$, the path deviation *falls* as a coarse grid under-samples the late turn.